

Visual Composition

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Image Generation Models

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt as a Contract

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Image generation and provenance. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Image generation and provenance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle. Evaluate this description.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

6. Solve the given case

Brief: large tree left, small bicycle right, open sky above. Output puts bicycle left.

Create a visual checklist.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 11 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create age-fit explanatory visuals with labeled purpose and no invented documentary evidence.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Visual Composition

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

Visual Composition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count objects

Two bottles and one beaker satisfy the count requirement.



2. Check position

Bottles on the right fail the required left-of relation.



3. Check direction

Arrow from beaker to bottles reverses the required flow.



4. Repair brief

Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

CHANGE ONE THING

Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Explain the changed result using the matching lesson.

Which checks pass after that single change, and what still needs repair?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

1. Check position
2. Repair brief
3. Count objects
4. Check direction

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output. Having every requested object is not the same as conveying the requested relationship. Position and arrow direction can change a scientific explanation while object counts remain correct. Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Practice 2: explain the deciding evidence

Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Position now passes; arrow direction still fails.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair a missing building block

Review Image Generation Models (ai-3/10). Goal: Explain the mechanism and terminology of image generation and provenance. Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured Output and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?. Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
